Course Name:	Digital Signal and Image Processing	Semester:	VI
Date of Performance:	02-03-2026	DIV/ Batch No:	C-3
Student Name:	Shreyans Tatiya	Roll No:	16010123325

Title: Write a program to apply the global processing technique: Histogram equalization. on a digital image

Objective: To learn and understand the concept of histogram stretching and equalization in image enhancement operations.

Expected Outcome of Experiment:

CO	Outcome
CO4	Design & implement algorithms for digital image enhancement, segmentation & restoration.

Books/ Journals/ Websites referred:

1. http://www.mathworks.com/support/
2. www.math.mtu.edu/~msgocken/intro/intro.html.
3. R. C.Gonsales R.E.Woods, “Digital Image Processing”, Second edition, Pearson Education
4. S.Jayaraman, S Esakkirajan, T Veerakumar “Digital Image Processing “Mc Graw Hill.
5. S.Sridhar,”Digital Image processing”, oxford university press, 1st edition."

Pre Lab/ Prior Concepts:

Image histogram:

In an image processing context, the histogram of an image normally refers to a histogram of the pixel intensity values. This histogram is a graph showing the number of pixels in an image at each different intensity value found in that image. For an 8-bit greyscale image there are 256 different possible intensities, and so the histogram will graphically display 256 numbers showing the distribution of pixels amongst those greyscale values.

Histograms can also be taken of color images either individual histogram of red, green and blue channels can be taken, or a 3-D histogram can be produced, with the three axes representing the red, blue and green channels, and brightness at each point representing the pixel count. The exact output from the operation depends upon the implementation it may simply be a picture of the required histogram in a suitable image format, or it may be a data file of some sort representing the histogram statistics.

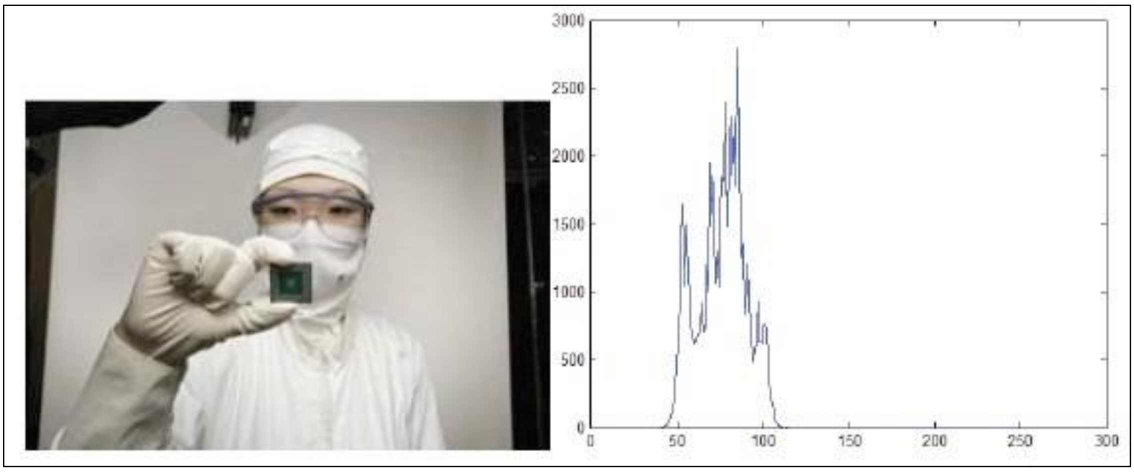


Fig. 1 An image and its histogram

Histogram Equalization:

A perfect image is one which has equal number of pixels in all its grey levels. hence our objective is not only to spread the dynamic range , but also to have equal pixels in all the grey levels. This technique is known as histogram equalization.

Basically the histogram equalization spreads out intensity values along the total range of values in order to achieve higher contrast. This method is especially useful when an image is represented by close contrast values, such as images in which both the background and foreground are bright at the same time, or else both are dark at the same time. For example, the result of applying histogram equalization to the image in figure 1 is presented in figure 2.

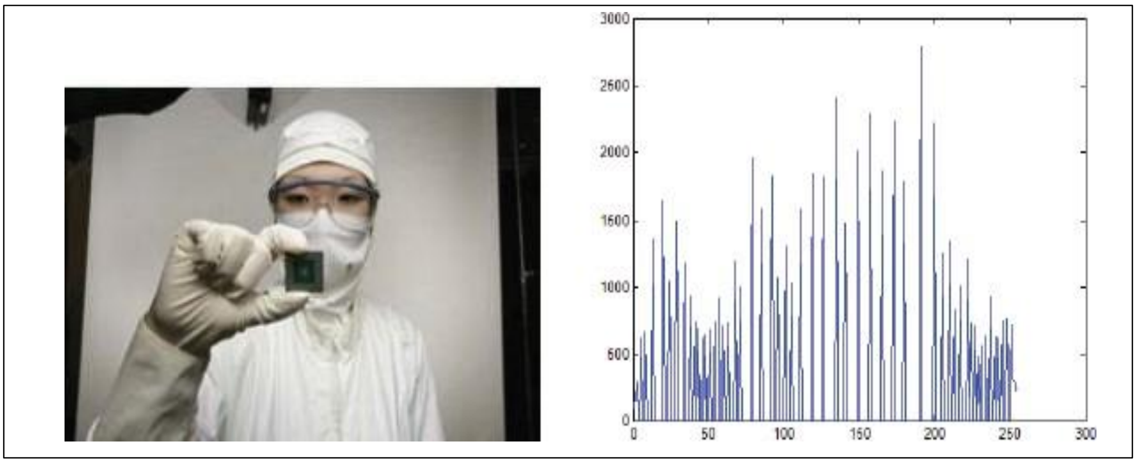


Fig. 2 New image and its equalized histogram

Description of cumulative histogram equalization:

Here are the steps for implementing this algorithm.

1. Create the histogram for the image.
2. Calculate the cumulative distribution function histogram.
3. Calculate the new values through the general histogram equalization formula.
4. Assign new values for each gray value in the image.

<u>Resources Used:</u> **MatLab**

Implementation steps with screenshots:

```
clc;
clear;
close all;

%% Read Image
I = imread('IMG_6232.jpg');

if size(I,3) == 3
    I = rgb2gray(I);
end

I = uint8(I);

L = 256;
max_gray_level = L-1;

%% Number of Pixels (Histogram)
num_pixels = zeros(1,L);

for k = 1:numel(I)
    gray = I(k);
    num_pixels(gray+1) = num_pixels(gray+1) + 1;
end

%% Running Sum
```

```
running_sum = zeros(1,L);
running_sum(1) = num_pixels(1);

for i = 2:L
    running_sum(i) = running_sum(i-1) + num_pixels(i);
end

%% Normalize
total_pixels = numel(I);
running_sum_norm = running_sum / total_pixels;

%% Multiply by Max Gray Level
multiplied_result = max_gray_level * running_sum_norm;

%% Round Values
transform_table = round(multiplied_result);

%% Apply Transformation
I_equalized = zeros(size(I),'uint8');

for i = 1:size(I,1)
    for j = 1:size(I,2)
        I_equalized(i,j) = transform_table(I(i,j)+1);
    end
end

%% Built-in Histogram Equalization
I_builtin = histeq(I);

%% Display Using Subplots

figure

subplot(3,2,1)
imshow(I)
title('Original Image')

subplot(3,2,2)
imhist(I)
title('Original Histogram')

subplot(3,2,3)
imshow(I_equalized)
title('Manual Equalization')

subplot(3,2,4)
imhist(I_equalized)
title('Manual Equalized Histogram')

subplot(3,2,5)
imshow(I_builtin)
title('Built-in histeq() Result')

subplot(3,2,6)
imhist(I_builtin)
title('Built-in Equalized Histogram')
```

Output

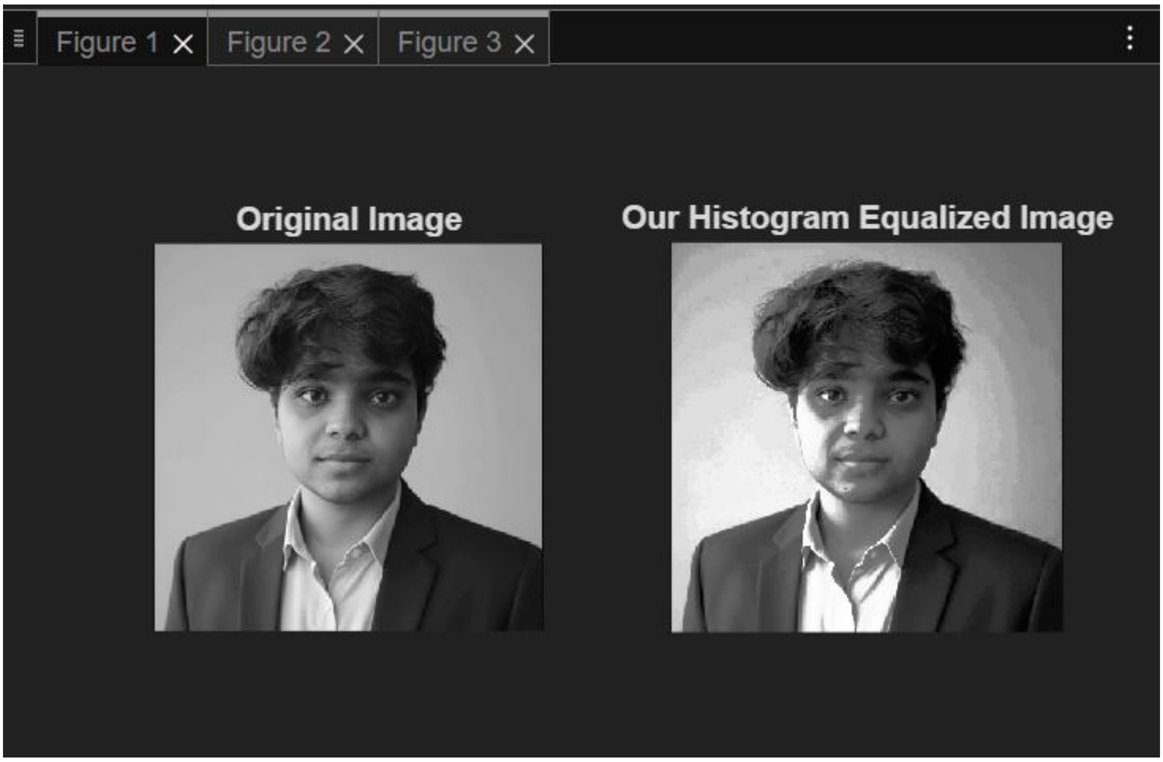


Figure 1
Figure 2
Figure 3
Original Image
Our Histogram Equalized Image

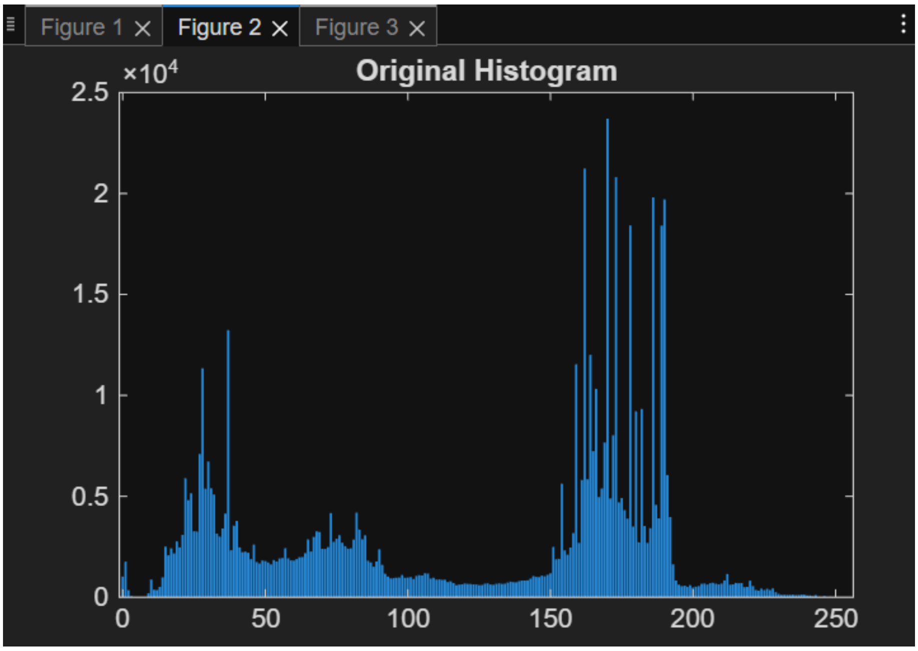
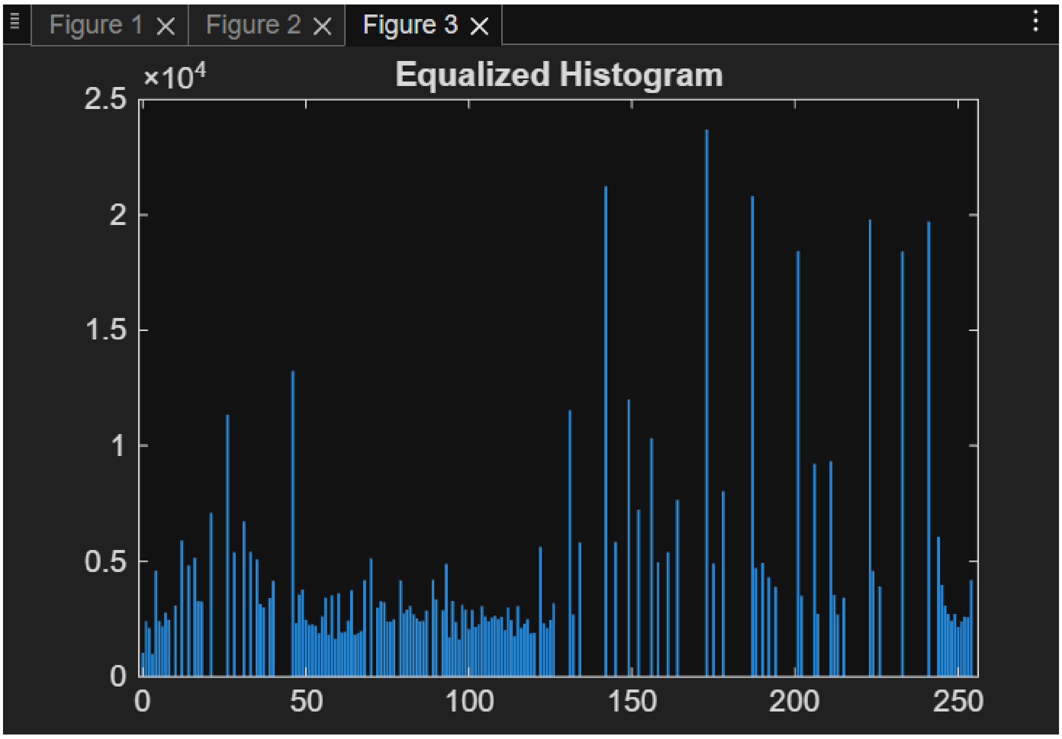


Figure 1
Figure 2
Figure 3
Original Histogram
×10^4
2.5
2
1.5
1
0.5
0
0
50
100
150
200
250



Conclusion:-

Histogram equalization improves the contrast of an image by redistributing pixel intensity values across the full range. It enhances visibility of details in both dark and bright regions, making the image clearer and easier to analyze.

Date: **Signature of faculty in-charge**

Post Lab Descriptive Questions

1. Compare between contrast stretching and histogram equalization.

Aspect	Contrast Stretching	Histogram Equalization
Definition	Expands intensity range of an image	Redistributes pixel intensities
Method	Linear transformation	Non-linear transformation (CDF-based)
Goal	Improve contrast by stretching min–max values	Enhance contrast by flattening histogram
Histogram Shape	Stretched but same shape	Becomes more uniform
Control	User-controlled (choose range)	Automatic
Effect	Mild enhancement	Strong enhancement
Use Case	When contrast is low but distribution is good	When image has poor contrast due to clustering

2. What is Histogram Equalization and what is its goal?

Histogram Equalization is an image processing technique that redistributes the intensity values of an image so that the histogram becomes approximately uniform.

Goal:

- Improve **contrast**
- Make hidden details more visible
- Spread out frequently occurring intensity values

3. What are the limitations of histogram equalization?

- **Over-enhancement:** Can make the image look unnatural
- **Noise Amplification:** Enhances noise along with details
- **Loss of Detail:** Fine details may get washed out
- **Not suitable for all images:** Especially poor for images with already good contrast

4. How can you interpret an image's contrast and brightness from its histogram?

You can understand an image just by looking at its histogram:

Brightness

- Histogram shifted **left** → Dark image
- Histogram shifted **right** → Bright image
- Centered → Normal brightness

Contrast

- **Narrow histogram** → Low contrast (flat image)
- **Wide spread histogram** → High contrast
- **Peaks in small range** → Poor detail in other regions

Additional Insights

- Gaps in histogram → Missing intensity levels
- Spikes → Dominance of certain intensity values